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(54) **DUAL RETRACTABLE PET LEASH WITH ABILITY TO SELECTIVELY CONTROL AND RESIST TANGLING OF LEASHES**

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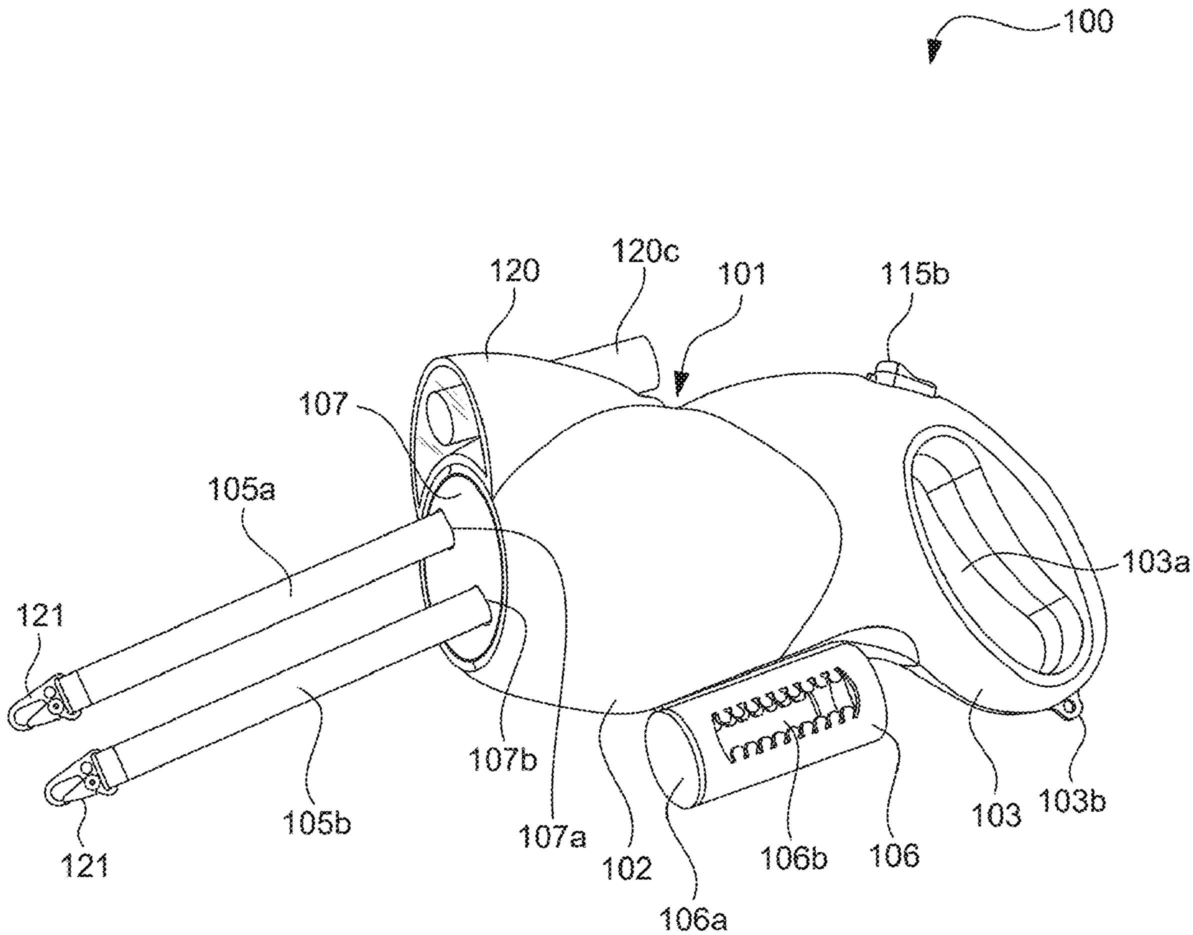
(57) **ABSTRACT**

A leash assembly for walking two pets is provided. The leash assembly includes a housing having a pair of spools to which leashes are secured. The housing includes a front panel rotatably connected to an outer tube. A right spool control tube is slidably nested within the left spool control tube, and the left spool control tube is slidably nested within the outer tube. The right spool control tube and left spool control tube is connected to a first control button and a second control button using brake rods. In operation, the control buttons, when pressed, pushes the spool control tubes forward, thereby disengaging locking protrusions of the spool control tubes engaged with one of notches of the spools. The control buttons, when released, retracts the right and left spool control tube backward, thereby reengaging the locking protrusions of the spool control tubes with one of the notches.



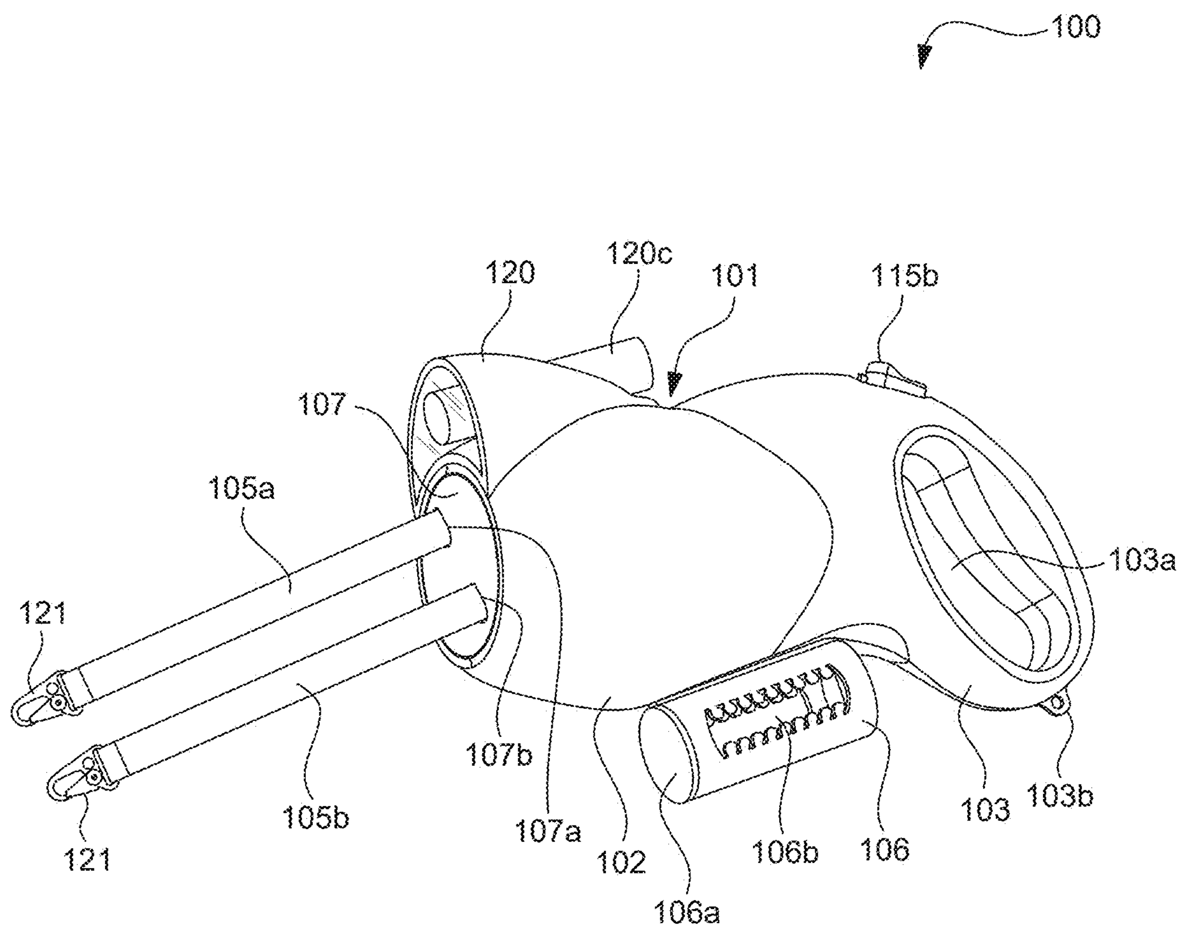
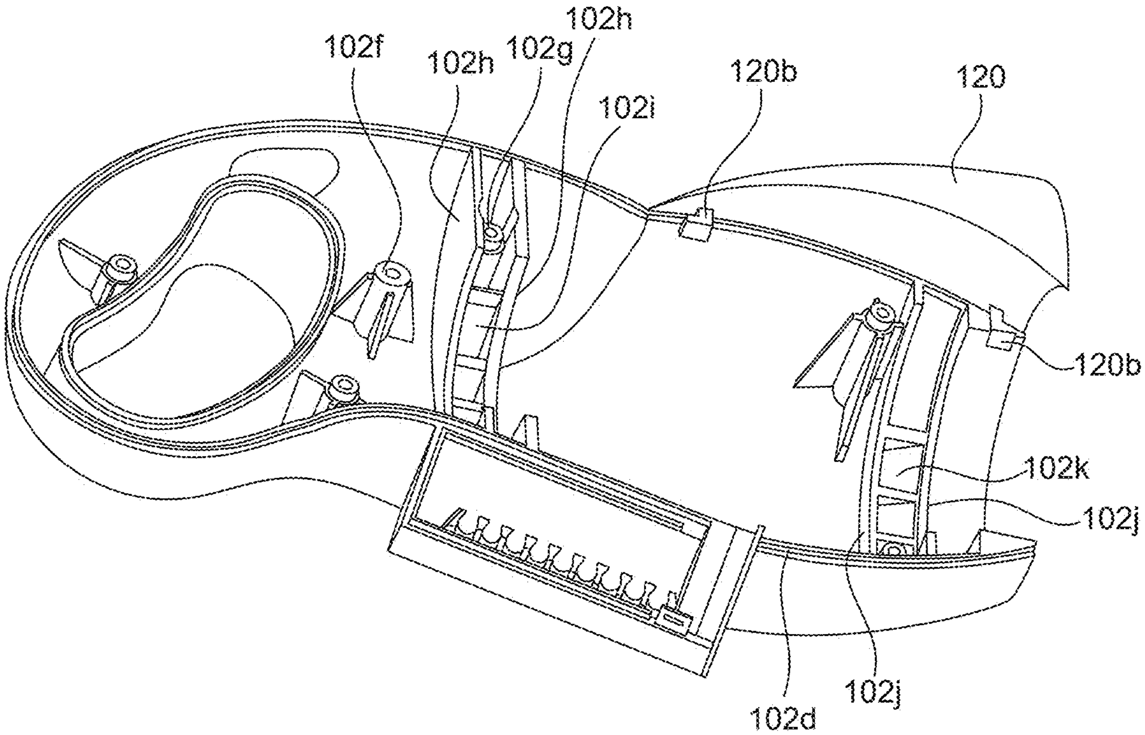

100
120
120c
101
115b
107
105a
103a
121
107b
105b
107a
102
106b
106
103
103b
121
106a

FIG. 1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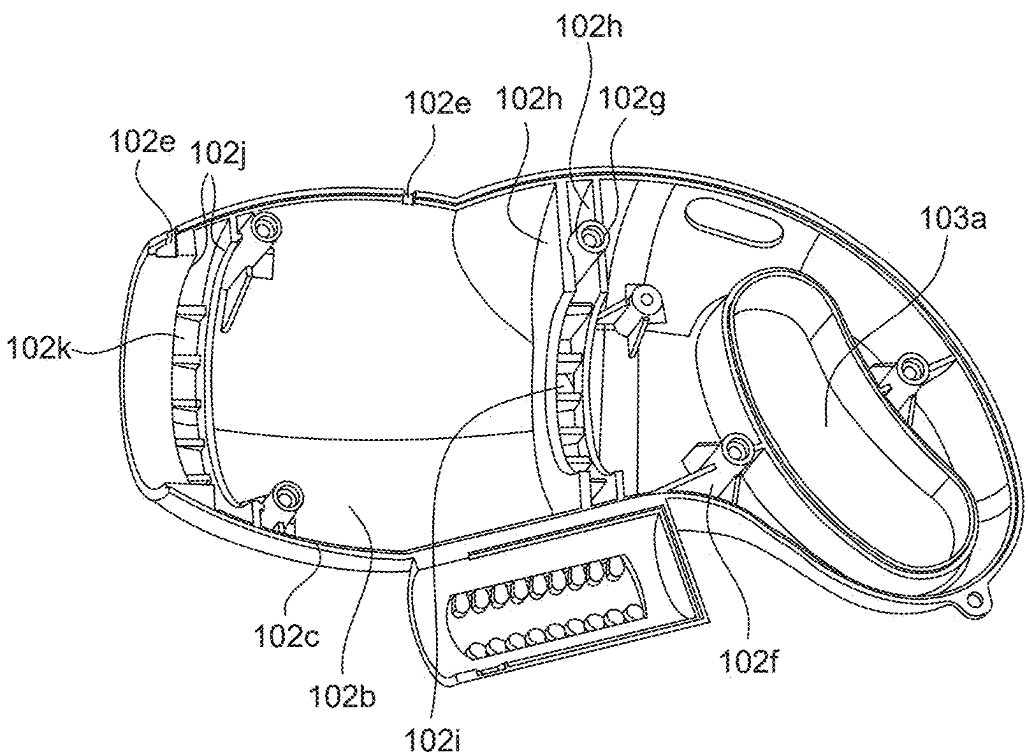
102g
102h
102i
120b
120
120b
102k
102j
102j
102d

FIG. 2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102g
102e
102j
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102b
102i
102f

FIG. 3

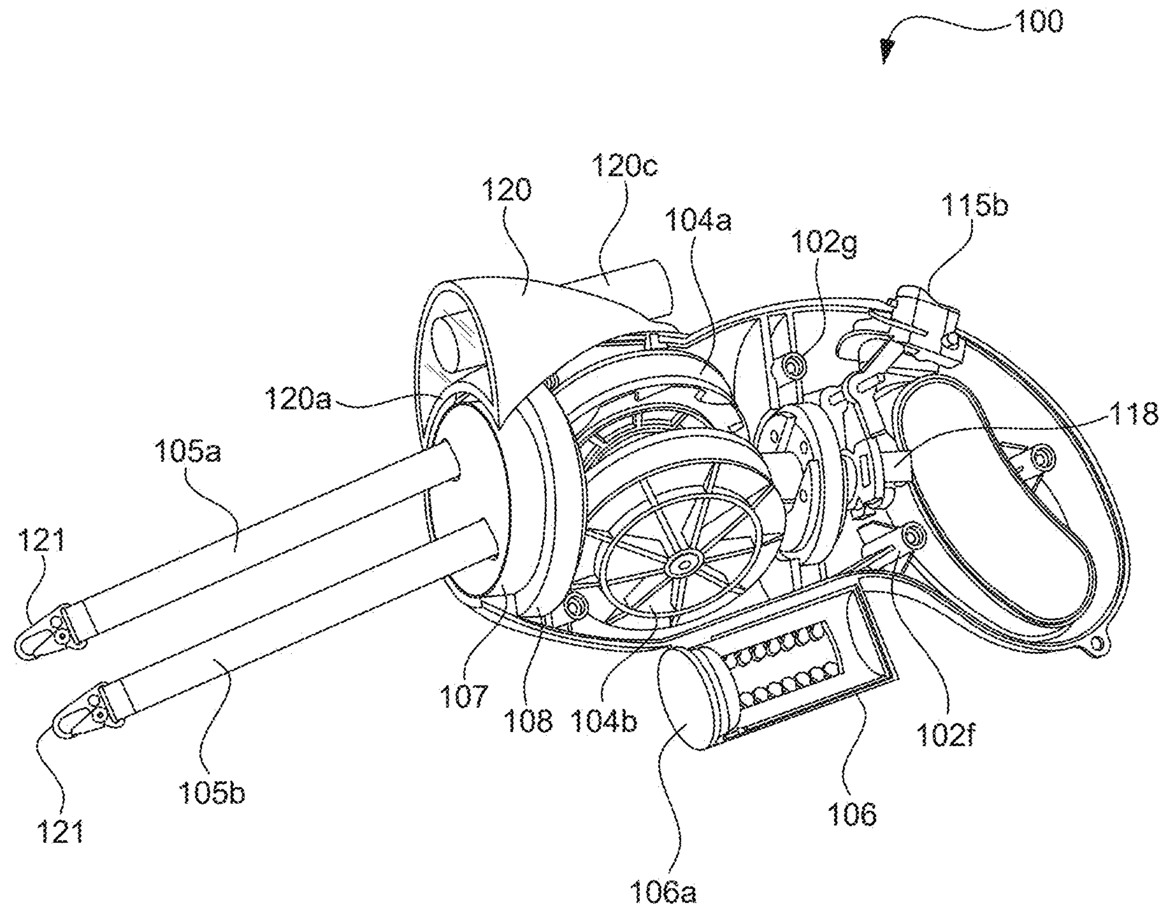
100
120c
120
104a
102g
115b
120a
118
105a
121
107
108
104b
102f
105b
106
121
106a

FIG. 4

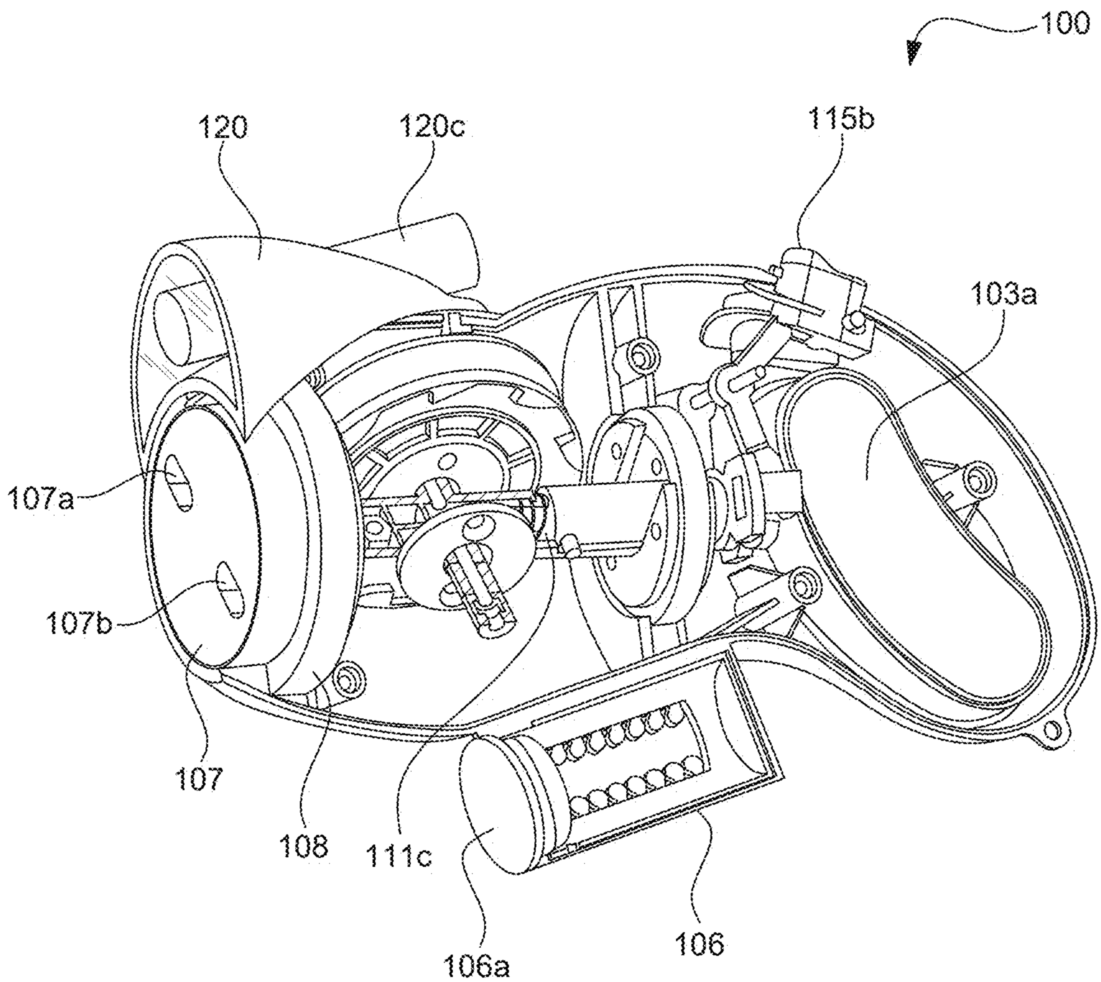


FIG. 5

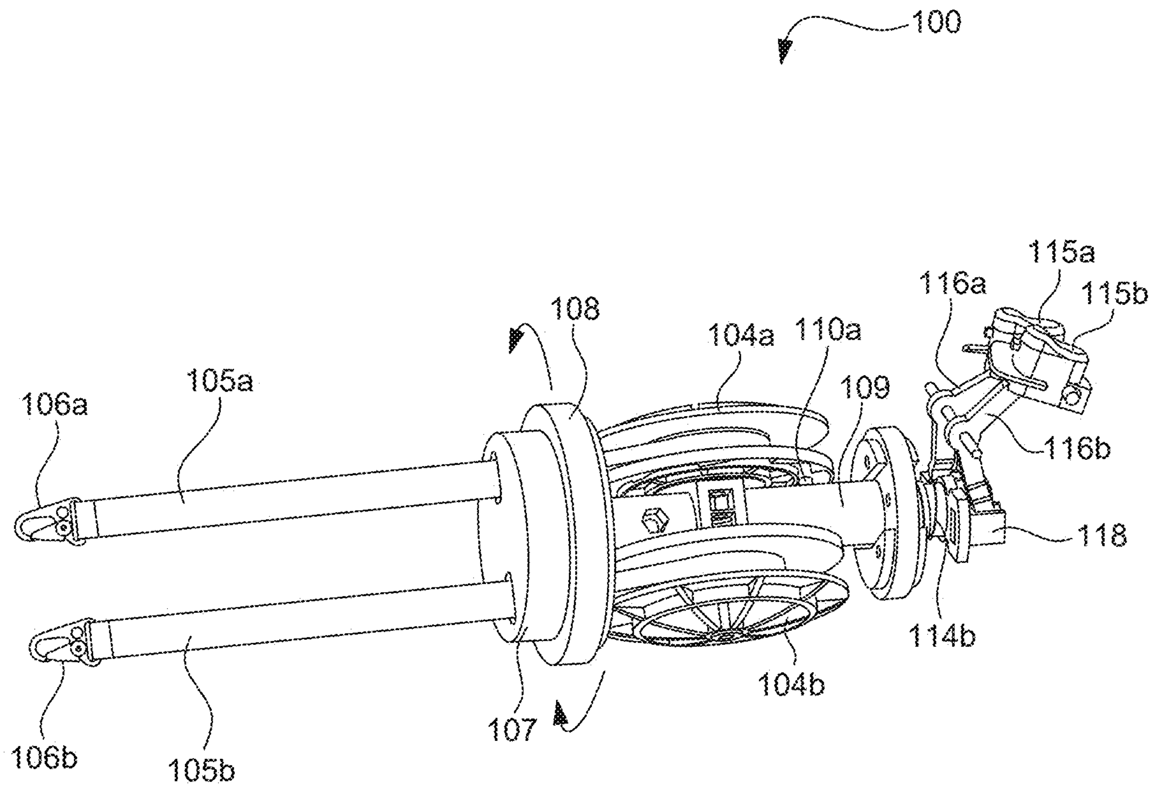
100
115a
116a
115b
108
104a
110a
109
106a
105a
116b
118
114b
104b
107
106b
105b

FIG. 6

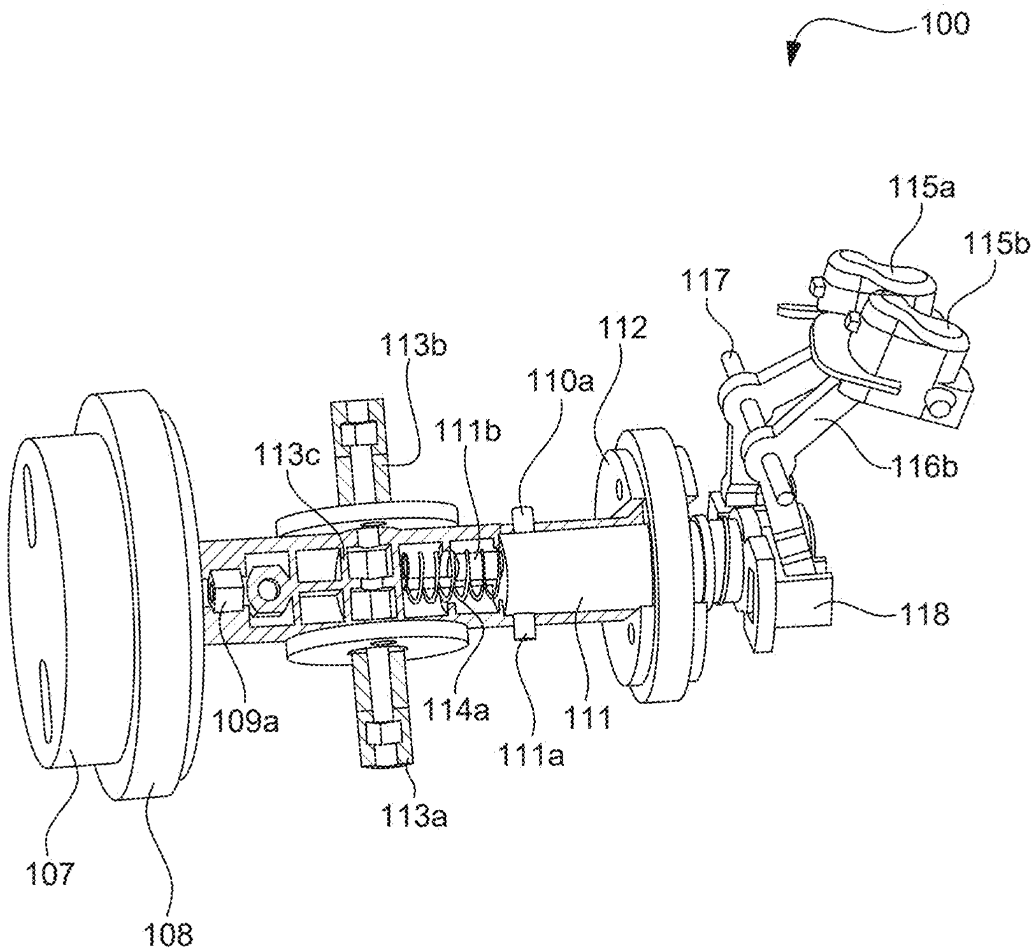
100
115a
115b
117
112
113b
110a
111b
113c
116b
118
109a
114a
111
111a
113a
107
108

FIG. 7

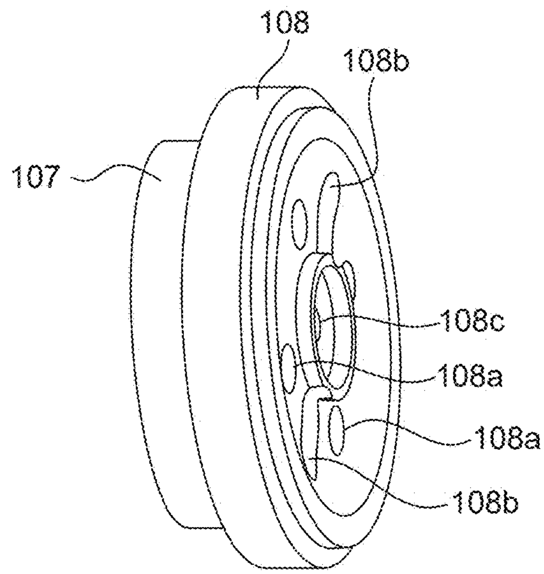


FIG. 8

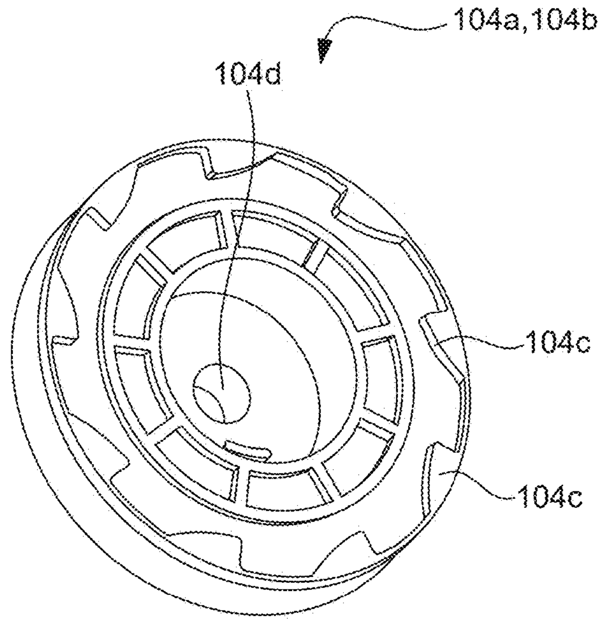


FIG. 9

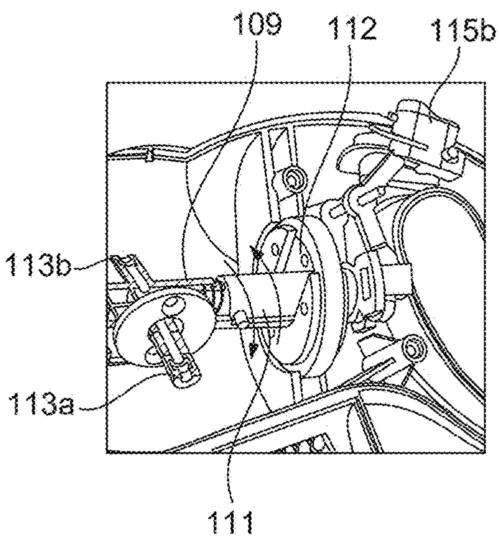


FIG. 10

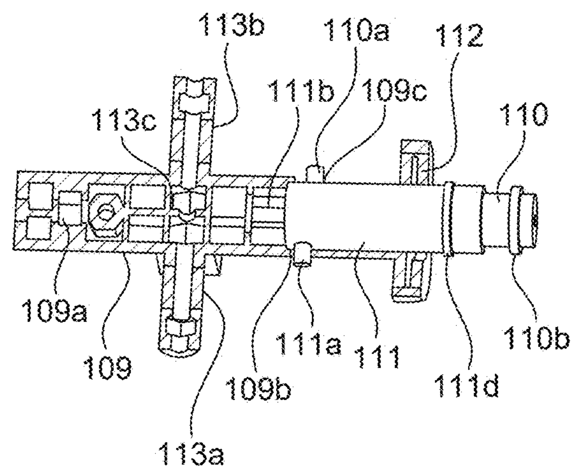


FIG. 11

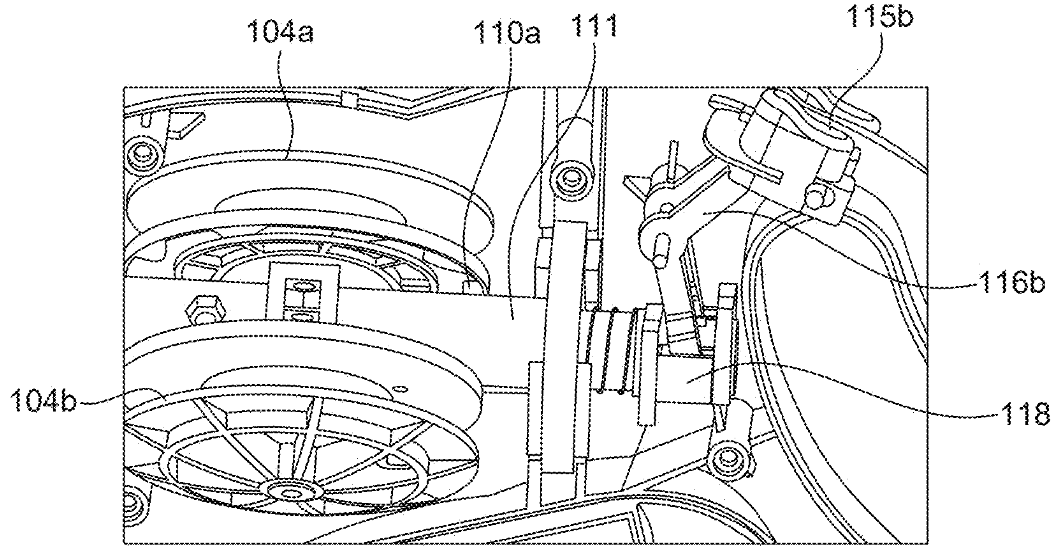
104a
110a
111
115b
116b
118
104b

FIG. 12

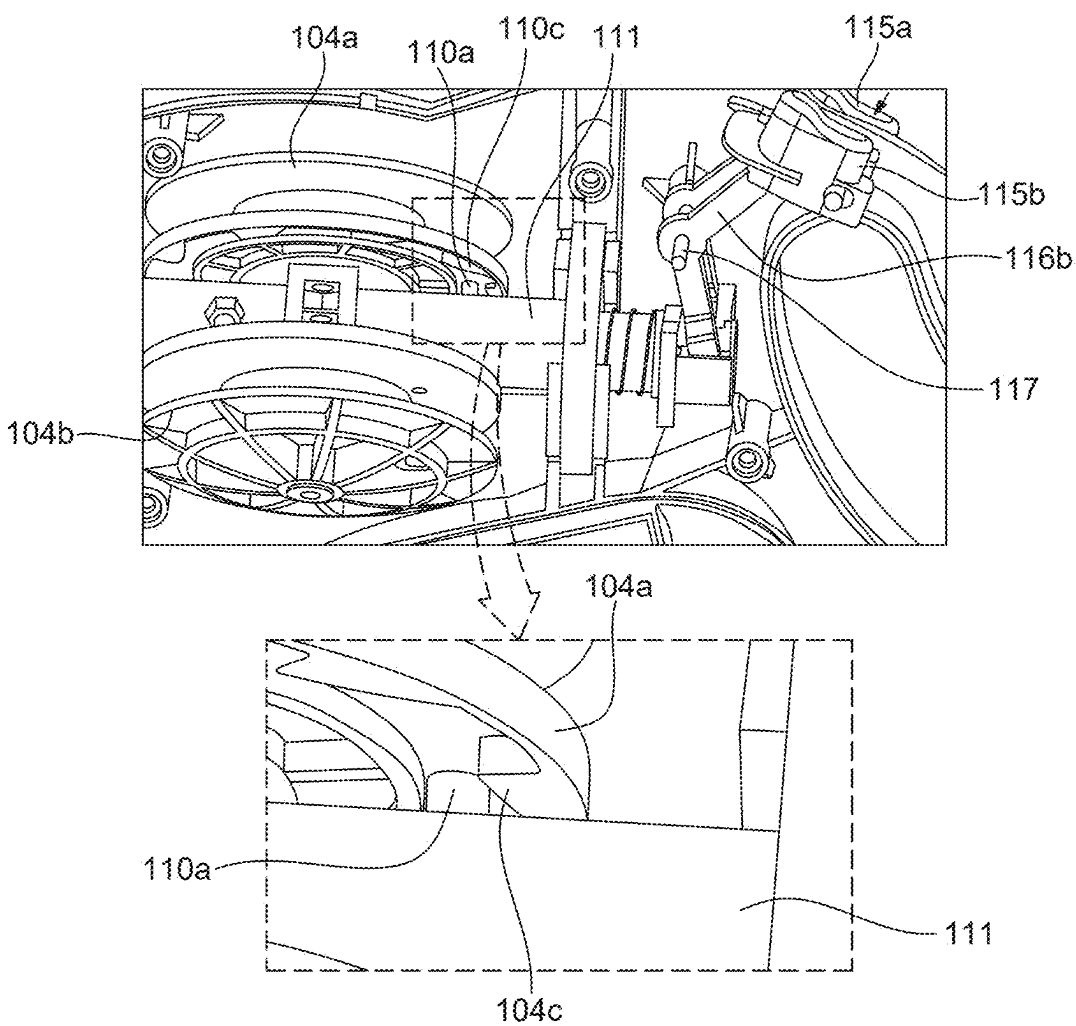
104a
110a
110c
111
115a
115b
116b
117
104b
104a
110a
111
104c

FIG. 13

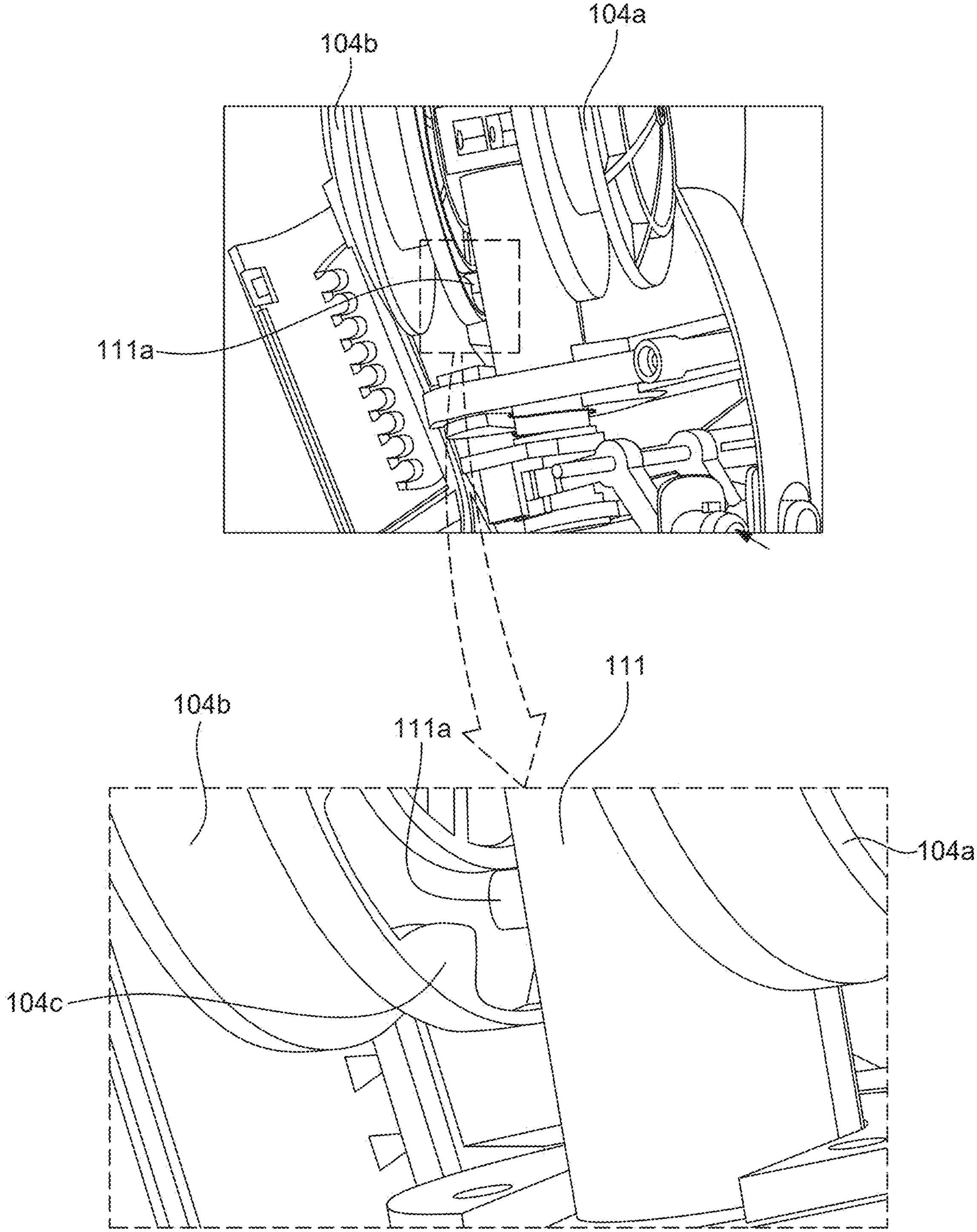


FIG. 14

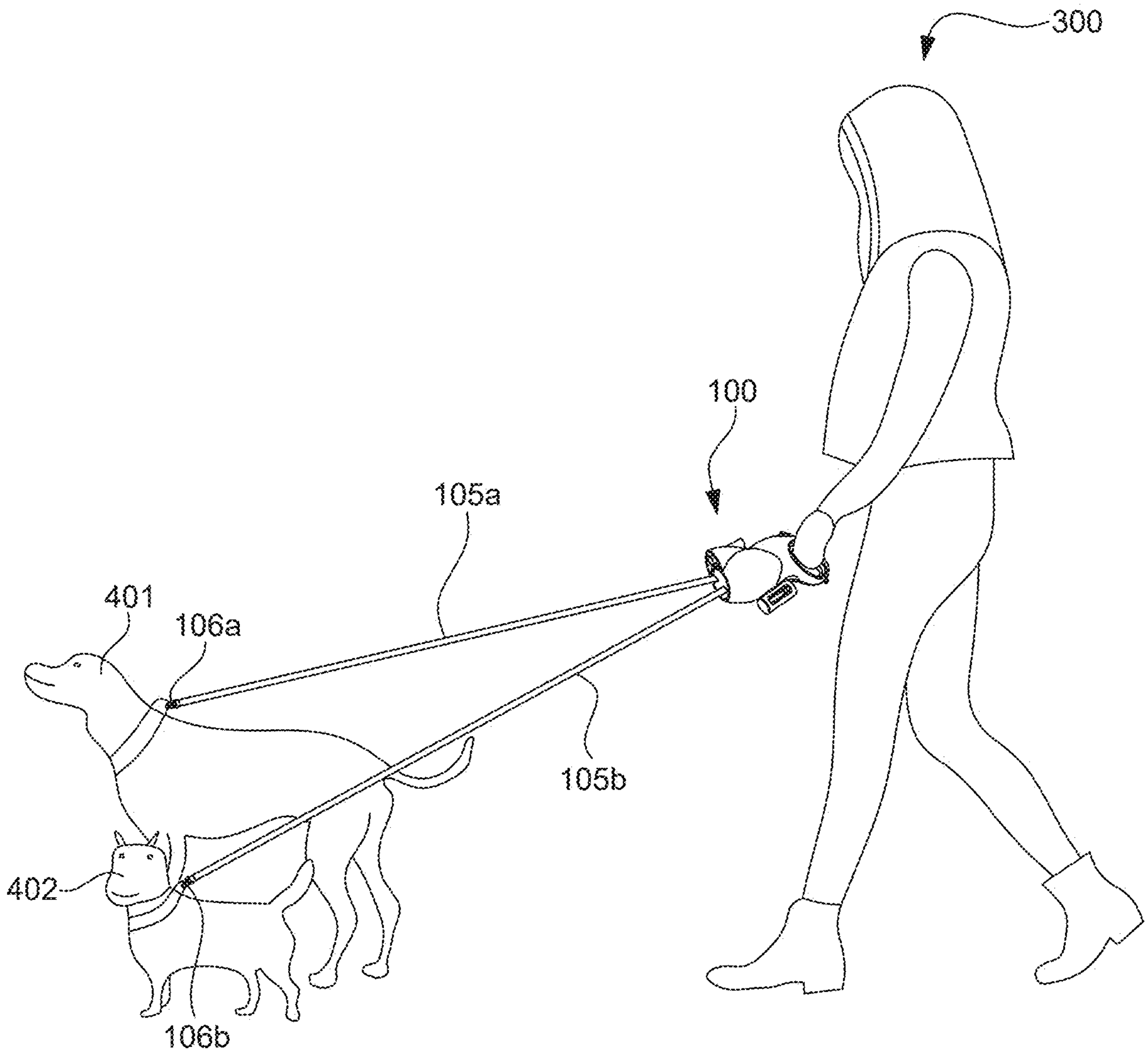


FIG. 15

# DUAL RETRACTABLE PET LEASH WITH ABILITY TO SELECTIVELY CONTROL AND RESIST TANGLING OF LEASHES

## FIELD OF THE INVENTION

**[0001]** The present invention relates to pet leashes, and more particularly to a dual-retractable pet leash used in connection with walking multiple pets simultaneously. Specifically, the present invention relates to a dual retractable pet leash with the ability to selectively or simultaneously control multiple pets, resist tangling of leashes, and minimize the effect each pet may have on each other and the owner holding the leash system in the natural course of the invention's use.

## BACKGROUND

**[0002]** Pets have become increasingly popular as companions for many people. However, just as people enjoy the companionship of pets, most animals also enjoy the companionship of humans. All pets require a certain amount of time and care on the part of the owner, with the time and care depending upon the species and size of the animal. In pets, many people own dogs, and in fact own multiple dogs. One of the necessities in owning such an animal is taking the dog for a walk. This usually entails the use of a leash, and when multiple dogs are owned, the owner must resort to either using more than one conventional leash or a leash specifically designed to handle more than one dog. While the owner may have an independent leash for each pet, this is somewhat redundant, the two pets will often attempt to act independently of one another in the outdoor environment. The owner of two pets who attempts to walk his or her pets using two leashes is often subjected to tugs and pulls in different directions as the owner attempts to control the two pets simultaneously.

**[0003]** Further, there exist leashes that help a pet owner to connect two pets and walk them; however, in such a leash system, a single strap flows out of the leash system that's then connected to two pets. When the pets try to walk or run independently then the two straps tied to the pets tangle. The pet owner then has to stop and manually remove the tangled straps. For example, Alcott ™ adventure retractable leashes sold on Amazon is a leash system with a single strap that winds or unwinds, and the single strap is connected to two dogs using an intermediary strap.

**[0004]** Additionally, many dual leash assemblies are known in the art, for example, the Dual Doggie leash assembly sold online at www.wigzi.com by Wigzi LLC of Washington D.C. is a dual leash having dual retractable leashes that spin relative to a handle to try to prevent the two leashes from tangling with each other. Freedom Leash, sold on ebay by Paws Republic is apparatus that discloses dual retractable leashes extending from a body that can spin to try to prevent the two leashes from tangling with each other.

**[0005]** U.S. Pat. No. 5,632,234 discloses a multiple pet leash assembly having a single leash running through a pulley with a handle to restrain the pulley and leash.

**[0006]** U.S. Pat. No. 9,016,242 discloses an apparatus for preventing the entanglement of a pair of lines fed from separate reels. U.S. Pat. No. 9,016,242 specifically discloses a multi-reel, line-handling apparatus that includes a housing, a shaft located within the housing, the shaft defining a shaft axis, a first reel located within the housing, the first reel rotatably mounted to the shaft about a first reel axis angled with respect to the shaft axis, and a second reel located within the housing, the second reel rotatably mounted to the shaft about a second reel axis angled with respect to the shaft axis. The first reel and the second reel can be rotatable with respect to the housing about the shaft axis. Each of the pets can be restrained independently by actuating the appropriate brake rod trigger. Actuating the inner brake rod trigger locks the first reel assembly, thus restraining the pet(s) on the first leash. Actuating the outer brake rod trigger locks the second reel assembly, thus restraining the pet(s) on the second leash. Each of the reel assemblies can be locked or unlocked independently. Locking one reel assembly has no effect on the other reel assembly and no effect on the rotation of the sub-assembly. Thus, either of the pets, all the pets, or neither pet, can be restrained while at the same time maintaining free rotation of the rotatable sub-assembly to permit automatic uncrossing of the leashes.

**[0007]** U.S. Patent Publication No. 20060201449 discloses a no-tangle two dog retractable leash with a flashlight. The device includes a housing containing a pair of spools, each having a leash secured thereto. The spools rotate about an axle to allow each leash to unwind from, or rewind onto, its respective spool. Both ends of the axle are attached to the housing, thereby fixing the axle in place inside the housing. A handle is attached to the housing by a shaft that allows the handle to pivot with respect to the housing about an axis extending transversely to the axle. Since the axle is fixed in place inside the housing, when one leash crosses over the other leash, the housing rotates with respect to the handle (held stationary in a user's hand) in order to uncross the leashes from one another. The rotation of the housing with respect to the handle in order to uncross the leashes detracts from the operation of the device. The no-tangle two dog retractable leash of US20060201449 also includes a brake pad that can be operated by a user by pressing a push pad. The brake pad is arranged to simultaneously engage both spools when the user presses the push pad, thereby stopping both spools simultaneously. This arrangement prevents the user from stopping one spool independently from the other.

**[0008]** Although several solutions in the form of dual leash apparatuses exist to prevent tangling of the two leashes during use, most of these apparatuses use complicated mechanisms for individual control of two spools with leashes for restraining the pets. These apparatuses usually employ two trigger buttons, each configured for controlling leashes on the spools. Upon selectively pressing these trigger buttons, the leashes are locked, preventing unwinding of the leashes, and depressing the buttons allows the leashes to unwind from the spools. In contrast, the present invention is directed towards a novel and more reliable dual retractable pet leash with the ability to selectively control unwinding or locking of the leashes and resist tangling of the two leashes.

## SUMMARY

**[0009]** It is an object of the present invention to provide a dual leash system with two control buttons, each control button configured to control the rotation of a spool carrying a leash. In the present invention, upon pressing a control button, the respective spool is unlocked to rotate and allow unwinding of the leash out of the unlocked spool.

**[0010]** It is an objective of the present invention to provide a dual retractable pet leash for walking or exercising two household pets (e.g., dogs), simultaneously.

**[0011]** It is another objective of the present invention to provide a dual retractable pet leash having an integral light.

**[0012]** It is another objective of the present invention to provide a dual retractable pet leash having an integral poop bag container.

**[0013]** It is another object of the present invention to provide a dual pet leash that is particularly adaptable for use with dogs, although it may be used with other animal species as well, for walking or otherwise restraining the pets as desired.

**[0014]** It is an object of the present invention to provide a dual retractable pet leash designed such that the leashes are individually controllable and automatically retract into the housing after the leashes are extended from the housing.

**[0015]** It is an object of the invention to provide a dual leash system with improved elements and arrangements thereof for the purposes described, which may be less complex, inexpensive, dependable, and fully effective in accomplishing its intended purposes.

**[0016]** Embodiments of the present invention discloses a dual-retractable pet leash assembly adapted for walking and restraining two pets. The dual-retractable pet leash assembly includes a housing comprising a front main body portion (**102**), and a rear handle portion with an opening to allow gripping of the rear handle portion.

**[0017]** In an embodiment, the housing (**101**) comprises a right spool, and a left spool rotatably disposed within the front main body portion of the housing. The right spool and the left spool having a first leash, a second leash secured thereto.

**[0018]** In an embodiment, the housing comprises a front panel integrated with a first bearing wheel rotatably connected to an outer tube residing within the housing.

**[0019]** In an embodiment, the housing includes a right spool control tube, and a left spool control tube. The right spool control tube slidably nested within the left spool control tube, and the left spool control tube is slidably nested within the outer tube. The right spool control tube is connected to a first control button using a first brake rod, and the left spool control tube is connected to the second control button using a second brake rod.

**[0020]** In an embodiment, the first control button, when pressed, pushes the right spool control tube forward, thereby disengaging a second locking protrusion of the right spool control tube engaged with one of a plurality of notches of the right spool, and the first control button, when released, retracts the right spool control tube backward, thereby reengaging the second locking protrusion of the right spool control tube with one of the plurality of notches.

**[0021]** In an embodiment, the second control button, when pressed, pushes the left spool control tube forward, thereby disengaging a first locking protrusion of the left spool control tube engaged with one of the plurality of notches of the left spool, and the second control button, when released, retracts the left spool control tube backward, thereby reengaging the first locking protrusion of the left spool control tube with one of the plurality of notches.

**[0022]** These and other features and advantages of the present invention will become apparent from the detailed description below, in light of the accompanying drawings.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0023]** The foregoing and other objects, features, and advantages of the invention will be apparent from the following more particular description of preferred embodiments of the invention, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention.

**[0024]** FIG. **1** is a dual retractable pet leash for walking two pets, according to an embodiment of the present invention.

**[0025]** FIG. **2** is a left section of a housing of the dual retractable pet leash of FIG. **1**, according to an embodiment.

**[0026]** FIG. **3** is a right section of a housing of the dual retractable pet leash of FIG. **1** without an integral light for clarity, according to an embodiment.

**[0027]** FIG. **4** is the dual retractable pet leash of FIG. **1** with the integral light and the left section removed, shown for clarity on internal components that reside within the housing, according to an embodiment.

**[0028]** FIG. **5** is the dual retractable pet leash of FIG. **4** with a cross-sectional view of a portion of an outer tube, and a left spool or reel removed, shown for clarity on how the spools are mounted on their respective axles, according to an embodiment.

**[0029]** FIG. **6** is a perspective view of the dual retractable pet leash of FIG. **1** taken without the housing, the integral light, the poop bag container and just showing the major components that reside within the housing as well as some components such as control buttons that reside outside the housing.

**[0030]** FIG. **7** is the dual retractable pet leash of FIG. **6** with a cross-sectional view of a portion of the outer tube, and the two spools mounted on their respective axles removed, shown for clarity of internal components and their configuration that extend within the outer tube, according to an embodiment.

**[0031]** FIG. **8** is a perspective view of a faceplate with an integrated bearing assembly through which the leashes pass, according to an embodiment.

**[0032]** FIG. **9** is a perspective view of an exemplary spool or reel mounted on axles configured on the sides of the outer tube located within the housing, according to an embodiment.

**[0033]** FIG. **10** is a sectional view of some of the major components that reside within the housing, specifically showing how detangling of the leashes is prevented in the natural course of the invention's use, according to an embodiment.

**[0034]** FIG. **11** is a partial sectional view of the outer tube with the two axles protruding from its sides, and a right spool control tube and a left spool control tube are shown nested within the outer tube.

**[0035]** FIG. **12** is a partial sectional view of major components configured with the housing, specifically showing the right spool in a locked state while the respective right control button remains depressed.

**[0036]** FIG. **13** is a partial sectional view of major components configured with the housing, specifically showing the right spool in an unlocked state while the respective right control button is pressed. A section of the figure is enlarged to show a shaft of the right spool control tube disengaged from the right spool.

**[0037]** FIG. **14** is a partial sectional view of major components configured with the housing, specifically showing the left spool in an unlocked state while the respective left control button is pressed. A section of the figure is enlarged to show a shaft of the left spool control tube disengaged from the left spool.

**[0038]** FIG. **15** shows a pet owner holding the dual leash system of FIG. **1** in the natural course of the invention's use.

## DETAILED DESCRIPTION

**[0039]** The following description is of exemplary embodiments of the invention only and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the following description is intended to provide a convenient illustration for implementing various embodiments of the invention. As will become apparent, various changes may be made in the function and arrangement of the elements described in these embodiments without departing from the scope of the invention as set forth herein. It should be appreciated that the description herein may be adapted to be employed with alternatively configured devices having different shapes, components, attachment mechanisms, and the like, and still fall within the scope of the present invention. Thus, the detailed description herein is presented for purposes of illustration only and not for limitation.

**[0040]** Reference in the specification to "one embodiment" or "an embodiment" is intended to indicate that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of the phrase "in one embodiment" or "an embodiment" in various places in the specification do not necessarily all refer to the same embodiment.

**[0041]** The dual-retractable pet leash and method of operating the same will now be described with reference to accompanying drawings, particularly FIGS. **1-15**.

**[0042]** Referring to FIGS. **1-4**, a dual-retractable pet leash **100** is provided. The dual-retractable pet leash **100** will hereinafter be referred to as a "leash assembly **100**". The dual-retractable pet leash **100** is ideally configured for walking two pets together (E.g. dogs). However, it should be understood that even though the leash assembly **100** is meant to walk and restrain two pets, it can be used for just walking a single pet as well.

**[0043]** The leash assembly **100** includes a housing **101**. The housing **101** comprises a front main body portion **102**, and a rear handle portion **103** with an opening **103***a* for a user's hand for gripping the leash assembly **100**. In the preferred embodiment, the front portion **102** and the rear handle portion **103** are integrally connected as a single unit, in the form of a single mold. In another embodiment (now shown), the front portion **102** and the rear handle portion **103** may be detachably connected using suitable connecting means such as notches. Further, the housing **101** may be formed using two sections (a left section **102***a*, and a right section **102***b* as shown in FIGS. **2-3**) that removably engage with each other. The two sections **102***a*, **102***b* are preferably symmetrical halves. The right section **102***b* and the left section **102***a* include interlocking features **102***c*,**102***d* along their edges to removably connect the two sections **102***a*, **102***b* together. The two sections **102***a*,**102***b* may comprise additional engaging provisions **102***f* for coupling the two sections together using screws. Although only one such provision is shown numbered in FIG. **2**, it should be understood that there may be 4-5 such provisions to tightly couple the two sections **102***a*,**102***b* together. Additionally, there may be one or more positioning pillars **102***g* configured on each of the sections **102***a*,**102***b* to reinforce the housing **101**. In another embodiment, the two sections **102***b*,**102***a* may be molded together as a single molded housing **101**. The housing **101** may be made using a suitable plastic, shock-proof, anti-crack, ABS material that may be rigid and durable.

**[0044]** In an embodiment, as seen in FIG. **1**, the dual-retractable pet leash **100** further comprises an integral light **120**. The light **120** is preferably an LED flashlight capable of illuminating several feet (E.g., 12 feet to 20 feet) distance for increasing visibility during nighttime walks. The flashlight **120** may include a press/depress button or any other mechanical button **120***c* to trigger the LED to switch on or off. The flashlight **120** may be removably attached to the housing **101**. Specifically, the flashlight **120** may be removably attached to the top of the front main body portion **102** of the housing **101**. The flashlight **120** may be attached on top of the front main body portion **102** using a snap-fit mechanism or any other similar mechanism that serves the purpose. The body of the flashlight **120** may be made using an 80-degree TPE soft rubber or any other suitable soft material. The bottom **120***a* of the flashlight **120** includes one or more locking features **120***b* that snap fit into one or more grooves **102***e* located on the front main body portion **102** of the housing **101** to lock the flashlight **120** over the front main portion **102**.

**[0045]** In an embodiment, as seen in FIG. **1**, the dual-retractable pet leash **100** further comprises an integral poop bag container **106**. The poop bag container **106** is configured for holding one or more pet poop bags (not shown). The user can make use of these poop bags while walking the pets for collection and disposal of pet's poop at appropriate location. The poop bag roll can be refilled in the container **106** by simply opening a lid **106***a* of the poop bag container **106**. The poop bag container **106** may be provided with at least one opening **106***b* to allow accessibility to the poop bags within the poop bag container **106**. The integral poop bag container **106** may be formed as an integral part of the housing **101** (as a single mold) located at the bottom of the housing **101**. In an alternative embodiment, the poop bag container **106** may be removably attached onto the housing **101** similar to the flashlight **104**. In an example, the poop bag container **106** may be snap fitted onto the housing **101**. The poop bag container **106** may be disposed at any other locations.

**[0046]** The housing **101** may further include a connecting tab **103***b*. The connecting tab **103***b* includes an opening to facilitate connecting various accessories such as for example, a retractable key chain for holding keys, any hanging tags and so on. The connecting tab **103***b* may be located at any suitable location on the housing **101**. In the example embodiment, the connecting tab **103***b* is shown configured at the bottom of the handle **103**, however, it may be possible that the connecting tab **103***b* may be configured at any other locations.

**[0047]** Referring to FIG. **4** in conjunction with FIGS. **5-7**, major components that reside within the housing **101**, as well as components such as control buttons that reside outside the housing **101** are shown. The description below will focus on various components that reside within the housing **101**, their interconnections, and functionality. As seen, the internal space within the housing **101** is kept sufficient to allow mounting of different components strategically within the housing **101**.

**[0048]** As seen, a front panel **107** integrated with a first bearing wheel **108** (also referred to as "front panel and first bearing wheel assembly **107**,**108**" is configured at a distal end of the housing **101**. The front panel **107** contains a first leash passage **107***a* and a second leash passage **107***b*, through which passes a first leash **105***a* and a second leash **105***b*. These leashes **105***a*,**105***b* can be made of any material useful for leashing pets and that can be wound around leash spools **104***a*,**104***b*. The leashes **105***a*,**105***b* may include a carabiner **106***a*,**106***b* or similar connectors connected to its free end that may couple to the clip or coupler located on the collar or harness of a pet. The bearing wheel **108** also contains similarly sized and appropriately aligned passages **108***b*. The leashes **105***a*,**105***b* wound around spools (a right spool **104***a*, and a left spool **104***b*) are inserted through bearing passages **108***b* located on the bearing wheel **108** and finally passed out of the first leash passage **107***a* and the second leash passage **107***b* for connecting them to pets intended for walking. The bearing wheel **108** is embodied within a groove **102***k* formed by a pair of first supporting members **102***j* to prevent dislodgement of the front panel **107** and the first bearing wheel **108** assembly. The front panel **107** is connected to the bearing wheel **108** using fasteners such as screws through the slots **108***a* located on the bearing wheel **108** as may be understood from FIG. **8**. The bearing wheel **108** spins within the groove **102***k* (which in turn spins/rotates the front panel **107** connected to the bearing wheel **108**), this allows the entire mechanism for locking/unlocking the spools **104***a*,**104***b* to rotate/spin internally (as indicated by arrows in FIG. **10**) when the two pets cross one another's paths while walking. This prevents the tangling of two leashes **105***a*,**105***b*. The rotational operation of the bearing wheel **108** is further supported by a second bearing wheel **112** located within the housing **101**. The second bearing wheel **112** is embodied within a groove **102***i* formed by a pair of second supporting members **102***h* to prevent dislodgement of the second bearing wheel **112** from its position.

**[0049]** The housing **101** further includes an outer tube **109**. The proximal end of the outer tube **109** is connected to the bearing wheel **108** using a fastener (not seen) that passes through a fastening hole **108***c* on the bearing wheel **108** and a hole **109***a* located inside the outer tube **109**. The outer tube **109** comprises a first slot **109***b* and a second slot **109***c* on its external surface.

**[0050]** A pair of spools, namely a right spool control tube **110** and a left spool control tube **111** are slidably nested within the outer tube **109**. In an embodiment, right the length of the spool control tube **110** is greater than the length of the left spool control tube **111**. The length of the outer tube **109** is ideally longer than the length of the right spool control tube **110** and the left spool control tube **111**. FIG. **11** more specifically shows this nesting of the right spool control tube **110** and the left spool control tube **111** within the outer tube **109**. As seen, the right spool control tube **110** is nested within the left spool control tube **111**, and the left spool control tube **111** is nested within the outer tube **109**. The outer tube **109** also includes a pair of axles, a first axle **113***a* and a second axle **113***b* extending outward from the outer tube **109**. The first axle **113***a* is adapted for rotatably mounting the left spool **104***b* thereon. The second axle **113***b* is adapted for rotatably mounting the right spool **104***a* thereon. Each of these axles **113***a*,**113***b* passes through a central hole **104***d* of the spools **104***a*,**104***b* and the spools **104***a*,**104***b* are retained in place using retaining means known in the art. Each of these spools **104***a*,**104***b* include a plurality of notches or teeth **104***c* (FIG. **9**). The notches **104***c* are seen configured in one particular pattern or orientation, however, it may be possible that these notches **104***c* may be configured in other patterns or orientations.

**[0051]** The left spool control tube **111** comprises a first locking protrusion **111***a*. The left spool control tube **111** may further comprise another protrusion **111***b* extending outward from its closed end **111***c*. The left spool control tube **111** further includes a first stopper **111***d*. The right spool control tube **110** comprises a second locking protrusion **110***a*, and a second stopper **110***b*. In assembly, when, the right spool control tube **110** is nested within the left spool control tube **111**, and the left spool control tube **111** is nested within the outer tube **109**, the protrusion **110***a* of the right spool control tube **110** emerges or extends out of the second slot **109***c*, and the protrusion **111***b* of the left spool control tube **111** emerges or extends out of the first slot **109***b*.

**[0052]** The right spool control tube **110** is connected to a first control button **115***a* using a first brake rod **116***a*. One end of the brake rod **116***a* is connected to the control button **115***a*, and the other end of the brake rod **116***a* is connected to a distal end of the right spool control tube **110** in proximity to the stopper **110***b* using a first connector **119**. The left spool control tube **111** is connected to a second control button **115***b* using a second brake rod **116***b*. One end of the brake rod **116***b* is connected to the control button **115***b*, and the other end of the brake rod **116***b* is connected to a distal end of the left spool control tube **111** in proximity to the stopper **111***d* using a second connector **118**. The break rods **116***a*,**116***b* are bent at an angle. In an embodiment, the break rods **116***a*,**116***b* may be bent at midway of their length. In another embodiment, the break rods **116***a*,**116***b* may be bent at any other locations along their length. The break rods **116***a*,**116***b* are connected via a shaft **117** that passes through both the break rods **116***a*,**116***b* at a location where the break rods **116***a*,**116***b* are bent. The shaft **117** linearly connects the two break rods **116***a*,**116***b*. The two break rods **116***a*,**116***b* rotate about the shaft **117** when the respective buttons **115***a*,**115***b* are pressed. In operation, each of the control buttons **115***a*,**115***b* are selectively pressed by the pet owner to control the release of leashes **105***a*,**105***b* from the dual retractable leash **100** during use. Upon pressing the button **115***a*,**115***b*, the corresponding break rods **116***a*,**116***b* rotate about the shaft **117** clockwise moving the right spool control tube **110** or the left spool control tube **111** forward, which disengages the protrusion **110***a* or the protrusion **111***a* of the tubes **110**,**111** from the notch **104***c* where the protrusions **110***a*,**111***a* are engaged. This unlocks the release of the corresponding leashes **105***a*,**105***b*. Next, upon releasing the button **115***a*,**115***b* (depressing), the corresponding break rods **116***a*,**116***b* rotate anticlockwise to regain their neutral positions or resting positions, moving the right spool control tube **110** or the left spool control tube **111** backward, which reengages the protrusion **110***a* or the protrusion **111***a* of the tubes **110**,**111** with one of the notches **104***c* (of the corresponding spools **104***a*,**104***b*).

**[0053]** The housing **101** further comprises a first spring **114***a* and a second spring **114***b*. The first spring **114***a* is disposed or extends between the closed end **111***c* of the left spool control tube **111** and an intermediate shaft **113***c* disposed inside the housing **101** along the same axis as that of the axles **113***a*,**113***b*. The second spring **114***b* is disposed or extends between a distal end of the outer tube **109** and the stopper **111***d* of the left spool control tube **111**. The protrusion **111***b* of the left spool control tube **111** is received within the first spring **114***a*. This protrusion **111***b* helps keep the left spool control tube **111** and nested tube **110** stable within the housing **101**. Further, there may be a third spring (not seen) disposed between a distal end of the left spool control tube **111** and the stopper **110***b* of the right spool control tube **110**. These springs help control buttons **115***a*,**115***b* regain their resting or neutral positions after a pet owner releases the buttons **115***a*,**115***b*. These springs also help the tubes **110**,**111** in making resistive movements forward when the pet owner presses the respective control buttons **115***a*,**115***b*.

**[0054]** The dual-retractable pet leash **100** may employ at least one battery (not seen in figures) useful for switching the flashlight **120** ON/OFF. The battery may be rechargeable battery.

**[0055]** Referring to FIGS. **12-15**, which specifically show and describe the use case scenario and operation of dual-retractable pet leash **100** while walking two dogs **401**,**402**. A pet owner **300** can hold the dual-retractable pet leash **100** using the handle **103**. While walking the two dogs **401**,**402**, the owner **300** ties the retractable leash **105***a*,**105***b* to pet dogs **401**,**402** (the carabiner **106***a*,**106***b* are attached to the collars of dogs as seen), and a distance between the dog and the owner can be controlled by releasing or retracting the leashes **105***a*,**105***b* with respect to the housing **101** according to a desired distance between the owner **300** and the pets **401**,**402**, so that the dogs **401**,**402** only move within a designated area around the owner **300**.

**[0056]** In operation, while walking the dogs **401**,**402**, when the dogs **401**,**402** try to cross each other, then the front panel **107** rotates, preventing tangling of the two leashes **105***a*,**105***b*. The entire tubular leash assembly that resides inside (discussed above) spins clockwise or anticlockwise depending upon which dog moves on which side. The rotation of the entire leash assembly (components inside the housing) inside is facilitated by the two bearing wheels **108**,**112** as described earlier in the description. This feature allows the entire mechanism inside the housing **101** to rotate preventing tangling of the leashes **105***a*,**105***b* that essentially minimizes the effect each pet may have on each other, and the owner **300** holding the leash assembly **100** in the natural course of the invention's use.

**[0057]** Further, when the owner **300** is just holding the leash assembly without pressing any of the control buttons **115***a*,**115***b*, the two leashes **105***a*,**105***b* wound around their respective spools **104***a*,**104***b* do not get released out of the spools **104***a*,**104***b* since the spools **104***a*,**104***b* remain in locked position as seen in FIG. **12**. This means if the dogs **401**,**402** try to run ahead demanding release of leashes **105***a*,**105***b*, the leashes **105***a*,**105***b* do not get released. If the owner of the pet **300** intends to release the leash **105***a* tied to the dog **401**, then the owner **300** needs to press the control button **115***a*. When the control button **115***a* is pressed, the breaking rod **116***a* rotates clockwise about the shaft **117** thereby pushing the right spool control tube **110** in a forward direction against the resistance of the spring. The right spool control tube **110**, when moved forward moves the protrusion **110***a* away from the notch **104***c* of the spool **104***a* (in a disengaged position), thereby unlocking the spool **104***a* as seen in FIG. **13**. When the right spool **104***a* is unlocked, the dog **401** can move free to move forward or pull the leash **105***a* forward (unwind the leash **105***a*) from the spool **105***a*. Again, when the owner releases the pressed button **115***a*, the right spool control tube **110** moves backward to its initial position reengaging the protrusion **110***a* with a notch **104***c* (one of the notches) of the right spool **104***a* (in an engaged position) as seen in FIG. **12**.

**[0058]** Likewise, if the owner **300** have to control release of leash **105***b* for the dog **402** then the owner has to press the control button **115***b* which in turn would rotate the breaking rod **116***b* in clockwise about the shaft **117** thereby pushing the left spool control tube **111** in a forward direction against the resistance of the spring **114***a*. The left spool control tube **111**, when moved forward, moves the protrusion **111***a* away from the notch **104***c* of the spool **104***b* (in a disengaged position), thereby unlocking the spool **104***b* as seen in FIG. **14**. When the left spool **104***b* is unlocked, the dog **402** can move free to move forward or pull the leash **105***b* forward (unwind the leash **105***b*) from the spool **105***b*. Again, when the owner releases the pressed button **115***b*, the left spool control tube **111** moves backward to its initial position reengaging the protrusion **111***a* with one of the notches **104***c* of the left spool **104***b* (in an engaged position) as seen in FIG. **12**. The owner **300** is facilitated to use the flashlight **120** provided on the housing **101**. The owner can push the button **120***c* to turn on the light and press again to switch off the light. The owner **300** is also provided with integrated poop bag container **106** to help the owner **300** to use the poop bags to collect dog's poop, if any of the dogs do poop while walking to keep the surrounding clean. The owner **300** is able to control the two dogs **401**,**402** as per his choice. Even if needed, the leash assembly **100** may be used by the owner to control just a single dog if he intends to walk a single dog. Further, the leash assembly **100** is designed in a way that allows the owner **300** to either individually control the leashing mechanism for either of two dogs or if the owner **300** wants, he can even simultaneously operate leashing for both the dogs by simultaneously pressing both the buttons **115***a*,**115***b*.

**[0059]** The housing **101** of the dual retractable leash assembly **100** as described above is formed using two symmetrical sections that removably engage with each other with the possibility of attaching the flashlight **120**, and the poop bag container **106** as an add on accessory. It should be understood that the housing **101** may be made as one single unit or mold (as described above). It is also possible that the housing may be made using a plurality of parts, say four parts. Likewise, the poop bag container **106** and the light **120** may be formed as a single mold with the housing instead of being external add on accessory as may be appreciated by those skilled in the art. The housing **101** may be formed in different aesthetic appearances and in different sizes using different materials. Further, the mechanism may further employ more or less number of springs inside for more precise and controlled movement of the tubes inside.

**[0060]** Finally, while the present invention has been described above with reference to various exemplary embodiments, many changes, combinations, and modifications may be made to the exemplary embodiments without departing from the scope of the present invention. For example, the various components may be implemented in alternative ways. These alternatives can be suitably selected depending upon the particular application or in consideration of any number of factors associated with the operation of the device. In addition, the techniques described herein may be extended or modified for use with other types of devices. These and other changes or modifications are intended to be included within the scope of the present invention.

What is claimed is:

1. A dual-retractable pet leash assembly (**100**) adapted for walking and restraining two pets (**401**,**401**), comprising:
a housing (**101**) comprising a front main body portion (**102**), and a rear handle portion (**103**) with an opening (**103***a*) to allow gripping of the rear handle portion (**103**), wherein the housing (**101**) comprises:
a right spool (**104***a*), and a left spool (**104***b*) rotatably disposed within the front main body portion (**102**) of the housing (**101**), wherein the right spool (**104***a*) and the left spool (**104***b*) having a first leash (**105***a*), a second leash (**105***b*) secured thereto;
a front panel (**107**) integrated with a first bearing wheel (**108**) rotatably connected to an outer tube (**109**) residing within the housing (**101**);
a right spool control tube (**110**), and a left spool control tube (**111**), wherein the right spool control tube (**110**) slidably nested within the left spool control tube (**111**), and the left spool control tube (**111**) is slidably nested within the outer tube (**109**);
wherein, the right spool control tube (**110**) is connected to a first control button (**115***a*) using a first brake rod (**116***a*), and the left spool control tube (**111**) is connected to the second control button (**115***a*) using a second brake rod (**116***a*);
wherein the first control button (**115***a*), when pressed, pushes the right spool control tube (**110**) forward, thereby disengaging a second locking protrusion (**110***a*) of the right spool control tube (**110**) engaged with one of a plurality of notches (**104***c*) of the right spool (**104***a*), and the first control button (**115***a*), when released, retracts the right spool control tube (**110**) backward, thereby reengaging the second locking protrusion (**110***a*) of the right spool control tube (**110**) with one of the plurality of notches (**104***c*); and
wherein the second control button (**115***b*), when pressed, pushes the left spool control tube (**111**) forward, thereby disengaging a first locking protrusion (**111***a*) of the left spool control tube (**111**) engaged with one of the plurality of notches (**104***c*) of the left spool (**104***b*), and the second control button (**115***b*), when released, retracts the left spool control tube (**111**) backward, thereby reengaging the first locking protrusion (**111***a*) of the left spool control tube (**111**) with one of the plurality of notches (**104***c*).

2. The dual-retractable pet leash (**100**) of claim **1** further comprising an integral light (**120**) mounted on the front main body portion (**102**) of the housing (**101**).

3. The dual-retractable pet leash (**100**) of claim **1** further comprising an integral poop bag container (**105**) configured for holding one or more pet poop bags.

4. The dual-retractable pet leash (**100**) of claim **1**, wherein the front panel (**107**) comprises a first leash passage (**107***a*) and a second leash passage (**107***b*), through which pass the first leash (**105***a*) and the second leash (**105***b*), respectively.

5. The dual-retractable pet leash (**100**) of claim **1**, wherein the right spool (**104***a*) and the left spool (**104***b*) are rotatably mounted on a first axle (**113***a*) and a second axle (**113***b*) extending outward from the outer tube (**109**).

6. The dual-retractable pet leash (**100**) of claim **1** further comprising a second bearing wheel (**112**) located within the housing (**101**) adapted for supporting the rotation of the front panel (**107**) and the first bearing wheel (**108**) assembly.

7. The dual-retractable pet leash (**100**) of claim **1**, wherein the first bearing wheel (**108**) is embodied within a first groove (**102***k*) formed by a pair of first supporting members (**102***j*), and the second bearing wheel (**112**) is embodied within a second groove (**102***i*) formed by a pair of second supporting members (**102***h*) to be stable during uses.

8. The dual-retractable pet leash (**100**) of claim **1**, wherein the right spool control tube (**110**) is connected to the first control button (**115***a*) using a first brake rod (**116***a*) and the left spool control tube (**111**) is connected to a second control button (**115***b*) using a second brake rod (**116***b*).

9. The dual-retractable pet leash (**100**) of claim **1**, wherein the first brake rod (**116***a*) and the second brake rod (**116***b*) are bent at an angle and are interconnected by a linear shaft (**117**) to rotate about the linear shaft (**117**).

10. The dual-retractable pet leash (**100**) of claim **1** further comprising a first spring (**114***a*) is disposed between a closed end (**111***c*) of the left spool control tube (**111**) and an intermediate shaft (**113***c*) disposed inside the housing (**101**) along the same axis as that of the first axle (**113***a*) and the second axle (**113***b*);
a second spring (**114***b*) is disposed between a distal end of the outer tube (**109**) and a first stopper (**111***d*) of the left spool control tube (**111**); and
a third spring (not seen) is disposed between a distal end of the left spool control tube (**111**) and the stopper (**110***b*) of the right spool control tube (**110**).

11. The dual-retractable pet leash (**100**) of claim **1**, wherein the first spring (**114***a*) receives a protrusion (**111***b*) extending outward from the closed end (**111***c*) of the left spool control tube (**111**).

12. The dual-retractable pet leash (**100**) of claim **1**, wherein the housing (**101**) is formed using a left section (**102***a*), and a right section (**102***b*) that removably engage with each other using their first interlocking feature (**102***d*), and a second interlocking feature (**102***c*), respectively.

13. The dual-retractable pet leash (**100**) of claim **2**, wherein a bottom (**120***a*) of the integral light (**120**) comprises one or more locking features (**120***b*) that snap fit into one or more grooves (**102***e*) located on the front main body portion (**102**) of the housing (**101**) to lock the integral light (**120**) over the front main portion (**102**).

14. The dual-retractable pet leash (**100**) of claim **1** further comprising a connecting tab (**103***b*) that includes an opening to facilitate connecting one or more accessories comprising at least a retractable key chain for holding keys, and a hanging tag.

15. The dual-retractable pet leash (**100**) of claim **1**, wherein the length of the right spool control tube (**110**) is greater than the length of the left spool control tube (**111**).

16. The dual-retractable pet leash (**100**) of claim **1**, wherein a proximal end of the outer tube (**109**) is connected to the first bearing wheel (**108**) using a fastener that passes through a fastening hole (**108***c*) located on the first bearing wheel (**108**) and a hole (**109***a*) located inside the outer tube (**109**).

17. The dual-retractable pet leash (**100**) of claim **1**, wherein the second locking protrusion (**110***a*) of the right spool control tube (**110**) extends out of a second slot (**109***c*) of the outer tube **109**, and the first locking protrusion (**111***a*)

of the left spool control tube (**111**_a_) extends out of the first slot (**109**_b_) of the outer tube (**109**_c_).

**18**. The dual-retractable pet leash (**100**) of claim **1**, wherein a first end of the first brake rod (**116**_a_) is connected to the first control button (**115**_a_), and a second end of the first brake rod (**116**_a_) is connected to a distal end of the right spool control tube (**110**) in proximity to a second stopper (**110**_b_) using a first connector (**119**).

**19**. The dual-retractable pet leash (**100**) of claim **1**, wherein a first end of the second brake rod (**116**_b_) is connected to the second control button (**115**_b_), and a second end of the second brake rod (**116**_b_) is connected to a distal end of the left spool control tube (**111**) in proximity to a first stopper (**111**_d_) using a second connector (**118**).

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